UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCES

INFORMATION TECHNOLOGY DEPARTMENT

SET **A**

FINAL EXAMINATION

Semester 1, Academic Year 2021 – 2022

MATH1200 - CALCULUS I

Day and Date : Monday, 27.12.2021 Time: 12:30 PM - 2:30 PM

Max. Marks	50	Time Allocation	2 hours
Student ID		Student Name	
Section		Level	Diploma

Instructions for students:

1. Exam paper consists of 8 pages including cover page. If any page is missing, inform the invigilator immediately.
2. Answer the questions in the space provided by using **blue or black** pen only.
3. Show the steps for the answers clearly.
4. Calculator is allowed.
5. Write the correct units, if any.
6. Any form of cheating is punishable. Students involved in cheating will be treated according to the Plagiarism and Academic Integrity Policies.

Question Type	**Total Marks**	**Marks Obtained**
PART A - Multiple Choice Questions	$10 \times 1 = 10$	
PART B - Short Answer Questions	$7 \times 4 = 28$	
PART C - Long Answer Question	$2 \times 6 = 12$	
Total Marks	50	

Marked By	**Checked/Re-marked by**	**Student Signature**

PART A: MULTIPLE CHOICE QUESTIONS

(Choose the correct answer. Each question carries 1 mark)

$(10 \times 1 = 10)$

1. Identify from the below table, the value of $\lim_{x \to 5} \frac{x^2 - 25}{x - 5}$ approaches to _____.

x	4.9	4.99	4.999	5.001	5.01	5.1
$f(x)$	9.9	9.99	9.999	10.001	10.01	10.1

(a) 10 (b) 9 (c) 5 (d) 4

2. $\lim_{x \to 3} (\ln e^{5x}) =$_____.

(a) e (b) 3 (c) 5 (d) 15

3. If the position of the particle is $s(t) = 3t^2 + t + 14$ *meters* then the instantaneous velocity of the particle at $t = 2$ *seconds* is _____.

(a) 28 m/sec (b) 6 m/sec (c) 13 m/sec (d) 14 m/sec

4. If $x = 5t$ and $y = 2t^5 - 5t$ then, $\frac{dy}{dx} =$ _____.

(a) $2t^4 - 1$ (b) $10t^4 - 1$ (c) $2t^4 + 1$ (d) $10t^4 + 1$

5. The radius of a circle decreases at a rate of 4 m/sec. When the radius is 6 m, the rate at which the area of a circle decreases is _____m^2/sec.

(a) -24π (b) 36π (c) 42π (d) 48π

6. The critical points of the function, $f(x) = 3x^3 - 36x$ are _____.

(a) $-2, \quad 2$ (b) $0, \quad 2$ (c) $-3, \quad 3$ (d) $-1, \quad 1$

7. The value of c by using mean value theorem, if $f(x) = x^2$ in the interval $[0, 2]$ is _____.

(a) 0 (b) 1 (c) 1.5 (d) 2

8. The inflection point of $f(x) = x^3 + 6x^2 + x + 10$ is _____.

(a) $(2, -24)$ (b) $(2, 24)$ (c) $(-2, 24)$ (d) $(-4, 10)$

9. The anti-derivative of the function $f(x) = 3x^2 + 10$ is _____.

(a) $3x^3 + 10x + c$ (b) $6x + c$ (c) $x^4 + 10x + c$ (d) $x^3 + 10x + c$

10. $\int \frac{1}{\csc x} dx =$ _____.

(a) $-\cos x + c$ (b) $\cos x + c$ (c) $\sin x + c$ (d) $-\sin x + c$

PART B: SHORT ANSWER QUESTIONS

(Answer all questions. Each question carries 4 marks)

$(7 \times 4 = 28)$

11. Determine the function $f(x) = \begin{cases} 4x+1, & x > 1 \\ 5, & x = 1 \\ -x+6, & x < 1 \end{cases}$ is continuous at $x = 1$. Justify the conclusion.

12. Find $\dfrac{dy}{dx}$, for $y = \ln(8x^3 + 19) + (1 + \tan^{-1} x)^4$.

13. The radius of a sphere increases at a rate of 5 m/sec. Find the rate at which the volume increases when the radius is 30 m.

14. Find the absolute maxima and absolute minima for the function $f(x) = 2x^2 + 4x + 6$ over $[-2, 5]$.

15. (a) The position of a hummingbird flying along a straight line in t *seconds* is given by

$s(t) = 4t^3 + 6t^2 + 1$ *meters*. Determine the acceleration of the bird at $t = 2$ *seconds*.

(b) Discuss the applicability of Rolle's theorem for the function $f(x) = x^{\frac{1}{3}}$ in $[-1, 1]$

16. Evaluate the definite integral $\int_0^2 (x+2)(x+3)\,dx$.

17. Find the indefinite integral, $\int \cos^{10}\theta \sin\theta \, d\theta$ by using substitution method. $(Hint : y = \cos\theta)$

PART C: LONG ANSWER QUESTIONS

(Answer all questions. Each question carries 6 marks)

$(2 \times 6 = 12)$

18. Find the equation of the tangent line to the graph of $3x^3 + 2x^2 + 8xy = 0$ at the point $(1, 2)$.

19. For $f(x) = x^3 - 12x + 10$, find the intervals where the function is increasing/decreasing, identify the local extreme values, concave up/ concave down, and the inflection point.

UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCES

INFORMATION TECHNOLOGY DEPARTMENT

SET **B**

FINAL EXAMINATION

Semester 1, Academic Year 2021 – 2022

MATH1200 - CALCULUS I

Day and Date : Monday, 27.12.2021 Time: 12:30 PM - 2:30 PM

Max. Marks	50	Time Allocation	2 hours
Student ID		Student Name	
Section		Level	Diploma

Instructions for students:

1. Exam paper consists of 8 pages including cover page. If any page is missing, inform the invigilator immediately.
2. Answer the questions in the space provided by using **blue or black** pen only.
3. Show the steps for the answers clearly.
4. Calculator is allowed.
5. Write the correct units, if any.
6. Any form of cheating is punishable. Students involved in cheating will be treated according to the Plagiarism and Academic Integrity Policies.

Question Type	**Total Marks**	**Marks Obtained**
PART A - Multiple Choice Questions	$10 \times 1 = 10$	
PART B - Short Answer Questions	$7 \times 4 = 28$	
PART C - Long Answer Question	$2 \times 6 = 12$	
Total Marks	50	

Marked By	**Checked/Re-marked by**	**Student Signature**

PART A: MULTIPLE CHOICE QUESTIONS

(Choose the correct answer. Each question carries 1 mark)

$(10 \times 1 = 10)$

1. Identify from the below table, the value of $\lim_{x \to 6} \frac{x^2 - 36}{x - 6}$ approaches to _____.

x	5.9	5.99	5.999	6.001	6.01	6.1
$f(x)$	11.9	11.99	11.999	12.001	12.01	12.1

(a) 6 (b) 5 (c) 11 (d) 12

2. $\lim_{x \to 3} (\ln e^{7x}) =$_____.

(a) 21 (b) e (c) 7 (d) 3

3. If the position of the particle is $s(t) = 5t^2 + t - 13$ *meters* then the instantaneous velocity of the particle at $t = 2$ *seconds* is _____.

(a) 13 m/sec (b) 21 m/sec (c) 9 m/sec (d) 10 m/sec

4. If $x = 6t$ and $y = 2t^6 + 6t$ then, $\frac{dy}{dx} =$ _____.

(a) $2t^4 - 1$ (b) $2t^5 + 1$ (c) $12t^5 + 6$ (d) $12t^5 - 6$

5. The radius of a circle decreases at a rate of 5 m/sec. When the radius is 7 m, the rate at which the area of a circle decreases is _____m^2/sec.

(a) -48π (b) 36π (c) 70π (d) 85π

6. The critical points of the function, $f(x) = 3x^3 - 81x$ are _____.

(a) 0, 2 (b) -1, 1 (c) -2, 2 (d) -3, 3

7. The value of c by using mean value theorem, if $f(x) = x^2$ in the interval $[0, 3]$ is _____.

(a) 1.5 (b) 0 (c) 2 (d) 1

8. The inflection point of $f(x) = x^3 - 6x^2 + x + 10$ is _____.

(a) $(-2, 4)$ (b) $(2, -4)$ (c) $(2, 24)$ (d) $(-4, 12)$

9. The anti-derivative of the function $f(x) = 4x^3 - 25$ is _____.

(a) $4x^3 + 25x + c$ (b) $x^4 + 25 + c$ (c) $x^4 - 25x + c$ (d) $12x^2 + c$

10. $\int \frac{1}{\sec x} dx =$ _____.

(a) $-\cos x + c$ (b) $\sin x + c$ (c) $\cos x + c$ (d) $-\sin x + c$

PART B: SHORT ANSWER QUESTIONS

(Answer all questions. Each question carries 4 marks)

$(7 \times 4 = 28)$

11. Determine the function $f(x) = \begin{cases} 5x+2, & x > 1 \\ 7, & x = 1 \\ -x+8, & x < 1 \end{cases}$ is continuous at $x = 1$. Justify the conclusion.

12. Find $\dfrac{dy}{dx}$, for $y = \ln(8x^4 - 20) + (6 + \tan^{-1} x)^5$.

13. The radius of a sphere increases at a rate of 7 m/sec. Find the rate at which the volume increases when the radius is 25 m.

14. Find the absolute maxima and absolute minima for the function $f(x) = 2x^2 - 4x + 5$ over $[-2, 5]$.

15. (a) The position of a hummingbird flying along a straight line in t *seconds* is given by

$s(t) = 4t^3 - 6t^2 + 1$ *meters*. Determine the acceleration of the bird at $t = 3$ *seconds*.

(b) Discuss the applicability of Rolle's theorem for the function $f(x) = x^{\frac{1}{3}}$ in $[-2, 2]$

16. Evaluate the definite integral $\int_0^3 (x+4)(x+5)\,dx$.

17. Find the indefinite integral, $\int \cos^{12}\theta \sin\theta\, d\theta$ by using substitution method. $(Hint : y = \cos\theta)$

PART C: LONG ANSWER QUESTIONS

(Answer all questions. Each question carries 6 marks)

$(2 \times 6 = 12)$

18. Find the equation of the tangent line to the graph $3x^3 - 2x^2 + 8xy = 0$ at the point $(1, 2)$.

19. For $f(x) = x^3 - 27x + 10$, find the intervals where the function is increasing/decreasing, identify the local extreme values, concave up/ concave down, and the inflection point.

UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCES

INFORMATION TECHNOLOGY DEPARTMENT

SET **A**

FINAL EXAMINATION

Semester 1, Academic Year 2021 – 2022

MATH1200 - CALCULUS I

Day and Date : Monday, 27.12.2021 Time: 12:30 PM - 2:30 PM

Max. Marks	50	Time Allocation	2 hours
Student ID		Student Name	
Section		Level	Diploma

Instructions for students:

1. Exam paper consists of 8 pages including cover page. If any page is missing, inform the invigilator immediately.
2. Answer the questions in the space provided by using **blue or black** pen only.
3. Show the steps for the answers clearly.
4. Calculator is allowed.
5. Write the correct units, if any.
6. Any form of cheating is punishable. Students involved in cheating will be treated according to the Plagiarism and Academic Integrity Policies.

Question Type	Total Marks	Marks Obtained
PART A - Multiple Choice Questions	$10 \times 1 = 10$	
PART B - Short Answer Questions	$7 \times 4 = 28$	
PART C - Long Answer Questions	$2 \times 6 = 12$	
Total Mark	50	

Exam Writer	Moderator	HoS

Answer Key

PART A: MULTIPLE CHOICE QUESTIONS

(Choose the correct answer. Each question carries 1 mark)

$(10 \times 1 = 10)$

1. Identify from the below table, the value of $\lim_{x \to 5} \frac{x^2 - 25}{x - 5}$ approaches to _____.

x	4.9	4.99	4.999	5.001	5.01	5.1
$f(x)$	9.9	9.99	9.999	10.001	10.01	10.1

(a) ✓10 (b) 9 (c) 5 (d) 4

2. $\lim_{x \to 3} (\ln e^{5x}) =$_____.

(a) e (b) 3 (c) 5 **(d) ✓15**

3. If the position of the particle is $s(t) = 3t^2 + t + 14$ *meters* then the instantaneous velocity of the particle at $t = 2$ *seconds* is _____.

(a) 28 m/sec (b) 6 m/sec **(c) ✓13 m/sec** (d) 14 m/sec

4. If $x = 5t$ and $y = 2t^5 - 5t$ then, $\frac{dy}{dx} =$ _____.

(a) ✓$2t^4 - 1$ (b) $10t^4 - 1$ (c) $2t^4 + 1$ (d) $10t^4 + 1$

5. The radius of a circle decreases at a rate of 4 m/sec. When the radius is 6 m, the rate at which the area of a circle decreases is _____m^2/sec.

(a) -24π (b) 36π (c) 42π **(d) ✓48π**

6. The critical points of the function, $f(x) = 3x^3 - 36x$ are _____.

(a) ✓ $-2, \quad 2$ (b) $0, \quad 2$ (c) $-3, \quad 3$ (d) $-1, \quad 1$

7. The value of c by using mean value theorem, if $f(x) = x^2$ in the interval $[0, 2]$ is _____.

(a) 0 **(b) ✓1** (c) 1.5 (d) 2

8. The inflection point of $f(x) = x^3 + 6x^2 + x + 10$ is _____.

(a) $(2, -24)$ (b) $(2, 24)$ **(c) ✓$(-2, 24)$** (d) $(-4, 10)$

9. The anti-derivative of the function $f(x) = 3x^2 + 10$ is _____.

(a) $3x^3 + 10x + c$ (b) $6x + c$ (c) $x^4 + 10x + c$ **(d) ✓$x^3 + 10x + c$**

10. $\int \frac{1}{\csc x} dx =$ _____.

(a) ✓ $-\cos x + c$ (b) $\cos x + c$ (c) $\sin x + c$ (d) $-\sin x + c$

Answer Key

PART B: SHORT ANSWER QUESTIONS

(Answer all questions. Each question carries 4 marks)

$(7 \times 4 = 28)$

11. Determine the function $f(x) = \begin{cases} 4x+1, & x > 1 \\ 5, & x = 1 \\ -x+6, & x < 1 \end{cases}$ is continuous at $x = 1$. Justify the conclusion.

Solution:

	$\because f(x)$ is continuous at $x = 1 \iff \lim_{x\to 1} f(x) = f(1)$
(1 mark)	$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (-x+6) = 5$
(1 mark)	$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (4x+1) = 5$
(1 mark)	Also, $f(1) = 5$
(0.5 marks)	$\therefore \lim_{x\to 1} f(x) = f(1) = 5$
(0.5 marks)	Hence, $f(x)$ is continuous at $x = 1$

12. Find $\dfrac{dy}{dx}$, for $y = \ln(8x^3+19) + (1+\tan^{-1} x)^4$.

Solution:

	Given function is $y = \ln(8x^3+19) + (1+\tan^{-1} x)^4 - (1)$
	Differentiating (1) with respect to x on both sides
(2 marks)	$\dfrac{dy}{dx} = \dfrac{1}{8x^3+19} \times \dfrac{d}{dx}(8x^3+19) + 4 \times (1+\tan^{-1} x)^3 \times \dfrac{d}{dx}(1+\tan^{-1} x)$
	Each term 1 Mark
(2 marks)	$\dfrac{dy}{dx} = \dfrac{1}{8x^3+19} \times (24x^2) + 4 \times (1+\tan^{-1} x)^3 \times \left(\dfrac{1}{1+x^2}\right)$
	Each term 1 Mark

Answer Key

13. The radius of a sphere increases at a rate of 5 m/sec. Find the rate at which the volume increases when the radius is 30 m.

Solution:

(0.5 marks)	Given $\frac{dr}{dt} = 5\ m/sec$ and $r = 30\ m$
(0.5 marks)	The volume of the sphere $v = \frac{4}{3}\pi r^3 - (1)$
	Differentiating (1) with respect to t on both sides
(1 mark)	$\frac{dv}{dt} = 4 \times \pi \times r^2 \times \frac{dr}{dt}$
(1 mark)	$\frac{dv}{dt} = 4 \times \pi \times (30)^2 \times 5$
(1 mark)	$\frac{dv}{dt} = 18000\pi\ m^3/sec$

14. Find the absolute maxima and absolute minima for the function $f(x) = 2x^2 + 4x + 6$ over $[-2, 5]$.

Solution:

	Given function is $f(x) = 2x^2 + 4x + 6$ over $[-2, 5]$
	Differentiating $f(x)$ with respect to x on both sides
(0.5 marks)	$f'(x) = 4x + 4$
(1 mark)	For critical points, make $f'(x) = 0$ then we get $x = -1 \in [-2, 5]$
(1 mark)	At the point $x = -1 \implies f(-1) = 4$, which is the absolute minimum value
(0.5 marks)	At the point $x = -2 \implies f(-2) = 6$
(1 mark)	At the point $x = 5 \implies f(5) = 76$, which is the absolute maximum value

Answer Key

15. (a) The position of a hummingbird flying along a straight line in t *seconds* is given by $s(t) = 4t^3 + 6t^2 + 1$ *meters*. Determine the acceleration of the bird at $t = 2$ *seconds*.

(b) Discuss the applicability of Rolle's theorem for the function $f(x) = x^{\frac{1}{3}}$ in $[-1, 1]$

Solution:

	Given position function is $s(t) = 4t^3 + 6t^2 + 1$
	Differentiating $s(t)$ with respect to t twice on both sides
(0.5 marks)	$v(t) = s^{'}(t) = 12t^2 + 12t$
(0.5 marks)	$a(t) = s^{''}(t) = 24t + 12$
(1 mark)	at $t = 2$ *sec*, the acceleration is, $a(2) = s^{''}(2) = 60\ m/sec^2$

Solution:

	Given function is $f(x) = x^{\frac{1}{3}}$ in $[-1, 1]$
(0.5 marks)	$f(x) = x^{\frac{1}{3}}$ is continuous on $[-1, 1]$
(0.5 marks)	$f(x)$ is not differentiable on $(-1, 1)$
(0.5 marks)	$\because$ at $x = 0 \implies f'(x) = \dfrac{1}{3x^{\frac{2}{3}}}$ is not defined
(0.5 marks)	Hence, Rolle's theorem is not applicable for the given function

Answer Key

16. Evaluate the definite integral $\int_0^2 (x+2)(x+3)\,dx$.

Solution:

(1 mark)	Given definite integral is $\int_0^2 (x+2)(x+3)\,dx = \int_0^2 (x^2+5x+6)\,dx$
(1 mark)	$= \left(\frac{x^3}{3} + 5\frac{x^2}{2} + 6x\right)_0^2$
(1 mark)	$= \left(\frac{2^3}{3} + 5 \times \frac{2^2}{2} + 6 \times 2\right) - \left(\frac{0^3}{3} + 5 \times \frac{0^2}{2} + 6 \times 0\right)$
(1 mark)	$= \frac{74}{3} = 24.6667$ units

17. Find the indefinite integral, $\int \cos^{10}\theta \sin\theta\, d\theta$ by using substitution method. $(Hint : y = \cos\theta)$

Solution:

	Given indefinite integral is $\int \cos^{10}\theta \sin\theta\, d\theta = \int (\cos\theta)^{10} \sin\theta\, d\theta - (1)$
(1 mark)	Let $y = \cos\theta$, then $-dy = \sin\theta d\theta$
	$\therefore$ equation $(1) \Longrightarrow$
(1 mark)	$\int \cos^{10}\theta \sin\theta\, d\theta = \int y^{10}(-dy)$
(1 mark)	$\int \cos^{10}\theta \sin\theta\, d\theta = -\frac{y^{11}}{11} + c$
(1 mark)	$\int \cos^{10}\theta \sin\theta\, d\theta = -\frac{(\cos\theta)^{11}}{11} + c$

Answer Key

PART C: LONG ANSWER QUESTIONS

(Answer all questions. Each question carries 6 marks)

$(2 \times 6 = 12)$

18. Find the equation of the tangent line to the graph of $3x^3 + 2x^2 + 8xy = 0$ at the point $(1, 2)$.

Solution:

	Given equation is $3x^3 + 2x^2 + 8xy = 0$—(1) and point=$(1, 2)$
	Differentiating (1) with respect to x on both sides
(0.5 marks)	$9x^2 + 4x + 8x\dfrac{dy}{dx} + 8y = 0$
(0.5 marks)	$8x\dfrac{dy}{dx} = -9x^2 - 4x - 8y$
(1 mark)	$\dfrac{dy}{dx} = \dfrac{-9x^2 - 4x - 8y}{8x}$
(1 mark)	$\left(\dfrac{dy}{dx}\right)_{at\ (1,2)} = \dfrac{-9x^2 - 4x - 8y}{8x}$
(1 mark)	$\left(\dfrac{dy}{dx}\right)_{at\ (1,2)} = \dfrac{-29}{8}$
	$\therefore$ The equation of the tangent line to the graph of the function at $(1, 2)$ is
(1 mark)	$y = \left(\dfrac{dy}{dx}\right)_{at\ (1,2)} \times (x - 1) + f(1)$
(1 mark)	$y = \dfrac{-29x + 45}{8}$

Answer Key

19. For $f(x) = x^3 - 12x + 10$, find the intervals where the function is increasing/decreasing, identify the local extreme values, concave up/ concave down, and the inflection point.

Solution:

	Given function is $f(x) = x^3 - 12x + 10 - (1)$
	Differentiating (1) with respect to x on both sides
(0.5 marks)	$f'(x) = 3x^2 - 12$
	$f'(x) = 0 \implies 3x^2 - 12 = 0$ then we get
(1 mark)	$x = \pm 2$ are the critical points.
(0.5 marks)	$f(x)$ is increasing in the intervals $(-\infty, -2)$ and $(2, \infty)$
(0.5 marks)	$f(x)$ is decreasing in the interval $(-2, 2)$
(0.5 marks)	at $x = -2$, local maximum occurs for the given function
(0.5 marks)	at $x = 2$, local minimum occurs for the given function
(0.5 marks)	$\because f'(x) = 3x^2 - 12$ then $f''(x) = 6x = 0 \implies x = 0$
(0.5 marks)	$f(x)$ is concave up in the interval $(0, \infty)$
(0.5 marks)	$f(x)$ is concave down in the interval $(-\infty, 0)$
(1 mark)	The inflection of point is $(0, 10)$

Answer Key

UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCES

INFORMATION TECHNOLOGY DEPARTMENT

SET **B**

FINAL EXAMINATION

Semester 1, Academic Year 2021 – 2022

MATH1200 - CALCULUS I

Day and Date : Monday, 27.12.2021 Time: 12:30 PM - 2:30 PM

Max. Marks	50	Time Allocation	2 hours
Student ID		Student Name	
Section		Level	Diploma

Instructions for students:

1. Exam paper consists of 8 pages including cover page. If any page is missing, inform the invigilator immediately.
2. Answer the questions in the space provided by using **blue or black** pen only.
3. Show the steps for the answers clearly.
4. Calculator is allowed.
5. Write the correct units, if any.
6. Any form of cheating is punishable. Students involved in cheating will be treated according to the Plagiarism and Academic Integrity Policies.

Question Type	**Total Marks**	**Marks Obtained**
PART A - Multiple Choice Questions	$10 \times 1 = 10$	
PART B - Short Answer Questions	$7 \times 4 = 28$	
PART C - Long Answer Questions	$2 \times 6 = 12$	
Total Mark	50	

Exam Writer	**Moderator**	**HoS**

Answer Key

PART A: MULTIPLE CHOICE QUESTIONS

(Choose the correct answer. Each question carries 1 mark)

$(10 \times 1 = 10)$

1. Identify from the below table, the value of $\lim_{x \to 6} \frac{x^2 - 36}{x - 6}$ approaches to _____.

x	5.9	5.99	5.999	6.001	6.01	6.1
$f(x)$	11.9	11.99	11.999	12.001	12.01	12.1

(a) 6 (b) 5 (c) 11 **(d) ✓12**

2. $\lim_{x \to 3} (\ln e^{7x}) =$_____.

(a) ✓21 (b) e (c) 7 (d) 3

3. If the position of the particle is $s(t) = 5t^2 + t - 13$ *meters* then the instantaneous velocity of the particle at $t = 2$ *seconds* is _____.

(a) 13 m/sec **(b) ✓21 m/sec** (c) 9 m/sec (d) 10 m/sec

4. If $x = 6t$ and $y = 2t^6 + 6t$ then, $\frac{dy}{dx} =$ _____.

(a) $2t^4 - 1$ **(b) ✓$2t^5 + 1$** (c) $12t^5 + 6$ (d) $12t^5 - 6$

5. The radius of a circle decreases at a rate of 5 m/sec. When the radius is 7 m, the rate at which the area of a circle decreases is _____m^2/sec.

(a) -48π (b) 36π **(c) ✓70π** (d) 85π

6. The critical points of the function, $f(x) = 3x^3 - 81x$ are _____.

(a) 0, 2 (b) -1, 1 (c) -2, 2 **(d) ✓ -3, 3**

7. The value of c by using mean value theorem, if $f(x) = x^2$ in the interval $[0, 3]$ is _____.

(a) ✓1.5 (b) 0 (c) 2 (d) 1

8. The inflection point of $f(x) = x^3 - 6x^2 + x + 10$ is _____.

(a) $(-2, 4)$ **(b) ✓$(2, -4)$** (c) $(2, 24)$ (d) $(-4, 12)$

9. The anti-derivative of the function $f(x) = 4x^3 - 25$ is _____.

(a) $4x^3 + 25x + c$ (b) $x^4 + 25 + c$ **(c) ✓$x^4 - 25x + c$** (d) $12x^2 + c$

10. $\int \frac{1}{\sec x} dx =$ _____.

(a) $-\cos x + c$ **(b) ✓ $\sin x + c$** (c) $\cos x + c$ (d) $-\sin x + c$

Answer Key

PART B: SHORT ANSWER QUESTIONS

(Answer all questions. Each question carries 4 marks)

$(7 \times 4 = 28)$

11. Determine the function $f(x) = \begin{cases} 5x+2, & x > 1 \\ 7, & x = 1 \\ -x+8, & x < 1 \end{cases}$ is continuous at $x = 1$. Justify the conclusion.

Solution:

	$\because f(x)$ is continuous at $x = 1 \iff \lim_{x\to 1} f(x) = f(1)$
(1 mark)	$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (-x+8) = 7$
(1 mark)	$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (5x+2) = 7$
(1 mark)	Also, $f(1) = 7$
(0.5 marks)	$\lim_{x\to 1} f(x) = f(1) = 7$
(0.5 marks)	Hence, $f(x)$ is continuous at $x = 1$

12. Find $\dfrac{dy}{dx}$, for $y = \ln(8x^4 - 20) + (6 + \tan^{-1} x)^5$.

Solution:

	Given function is $y = \ln(8x^4 - 20) + (6 + \tan^{-1} x)^5 - (1)$
	Differentiating (1) with respect to x on both sides
(2 marks)	$\dfrac{dy}{dx} = \dfrac{1}{8x^4 - 20} \times \dfrac{d}{dx}(8x^4 - 20) + 5 \times (6 + \tan^{-1} x)^4 \times \dfrac{d}{dx}(6 + \tan^{-1} x)$
	Each term 1 Mark
(2 marks)	$\dfrac{dy}{dx} = \dfrac{1}{8x^4 - 20} \times (32x^3) + 5 \times (6 + \tan^{-1} x)^4 \times \left(\dfrac{1}{1 + x^2}\right)$
	Each term 1 Mark

Answer Key

13. The radius of a sphere increases at a rate of 7 m/sec. Find the rate at which the volume increases when the radius is 25 m.

Solution:

(0.5 marks)	Given $\frac{dr}{dt} = 7\ m/sec$ and $r = 25\ m$
(0.5 marks)	The volume of the sphere $v = \frac{4}{3}\pi r^3 - (1)$
	Differentiating (1) with respect to t on both sides
(1 mark)	$\frac{dv}{dt} = 4 \times \pi \times r^2 \times \frac{dr}{dt}$
(1 mark)	$\frac{dv}{dt} = 4 \times \pi \times (25)^2 \times 7$
(1 mark)	$\frac{dv}{dt} = 17500\pi\ m^3/sec$

14. Find the absolute maxima and absolute minima for the function $f(x) = 2x^2 - 4x + 5$ over $[-2, 5]$.

Solution:

	Given function is $f(x) = 2x^2 - 4x + 5$ over $[-2, 5]$
	Differentiating $f(x)$ with respect to x on both sides
(0.5 marks)	$f'(x) = 4x - 4$
(1 mark)	For critical points, make $f'(x) = 0$ then we get $x = 1 \in [-2, 5]$
(1 mark)	At the point $x = 1 \implies f(1) = 3$, which is the absolute minimum value
(0.5 marks)	At the point $x = -2 \implies f(-2) = 21$
(1 mark)	At the point $x = 5 \implies f(5) = 35$, which is the absolute maximum value

Answer Key

15. (a) The position of a hummingbird flying along a straight line in t *seconds* is given by $s(t) = 4t^3 - 6t^2 + 1$ *meters*. Determine the acceleration of the bird at $t = 3$ *seconds*.

(b) Discuss the applicability of Rolle's theorem for the function $f(x) = x^{\frac{1}{3}}$ in $[-2, 2]$

Solution:

	Given position function is $s(t) = 4t^3 - 6t^2 + 1$
	Differentiating $s(t)$ with respect to t twice on both sides
(0.5 marks)	$v(t) = s^{'}(t) = 12t^2 - 12t$
(0.5 marks)	$a(t) = s^{''}(t) = 24t - 12$
(1 mark)	At $t = 3$ *sec*, the acceleration is, $a(3) = s^{''}(3) = 60\ m/sec^2$

Solution:

	Given function is $f(x) = x^{\frac{1}{3}}$ in $[-2, 2]$
(0.5 marks)	$f(x) = x^{\frac{1}{3}}$ is continuous on $[-2, 2]$
(0.5 marks)	$f(x)$ is not differentiable on $(-2, 2)$
(0.5 marks)	$\because$ at $x = 0 \implies f'(x) = \dfrac{1}{3x^{\frac{2}{3}}}$ is not defined
(0.5 marks)	Hence, Rolle's theorem is not applicable for the given function

Answer Key

16. Evaluate the definite integral $\int_0^3 (x+4)(x+5)\,dx$.

Solution:

(1 mark)	Given definite integral is $\int_0^3 (x+4)(x+5)\,dx = \int_0^3 (x^2+9x+20)\,dx$
(1 mark)	$= \left(\frac{x^3}{3} + 9\frac{x^2}{2} + 20x\right)_0^3$
(1 mark)	$= \left(\frac{3^3}{3} + 9 \times \frac{3^2}{2} + 20 \times 3\right) - \left(\frac{0^3}{3} + 9 \times \frac{0^2}{2} + 20 \times 0\right)$
(1 mark)	$= \frac{219}{3} = 109.5$ units

17. Find the indefinite integral, $\int \cos^{12}\theta \sin\theta\, d\theta$ by using substitution method. $(Hint : y = \cos\theta)$

Solution:

	Given indefinite integral is $\int \cos^{12}\theta \sin\theta\, d\theta = \int (\cos\theta)^{12} \sin\theta\, d\theta - (1)$
(1 mark)	Let $y = \cos\theta$, then $-dy = \sin\theta d\theta$
	$\therefore$ equation $(1) \Longrightarrow$
(1 mark)	$\int \cos^{12}\theta \sin\theta\, d\theta = \int y^{12}(-dy)$
(1 mark)	$\int \cos^{12}\theta \sin\theta\, d\theta = -\frac{y^{13}}{13} + c$
(1 mark)	$\int \cos^{12}\theta \sin\theta\, d\theta = -\frac{(\cos\theta)^{13}}{13} + c$

Answer Key

PART C: LONG ANSWER QUESTIONS

(Answer all questions. Each question carries 6 marks)

$(2 \times 6 = 12)$

18. Find the equation of the tangent line to the graph $3x^3 - 2x^2 + 8xy = 0$ at the point $(1, 2)$.

Solution:

	Given equation is $3x^3 - 2x^2 + 8xy = 0$—(1) and point=$(1, 2)$
	Differentiating (1) with respect to x on both sides
(0.5 marks)	$9x^2 - 4x + 8x\dfrac{dy}{dx} + 8y = 0$
(0.5 marks)	$8x\dfrac{dy}{dx} = -9x^2 + 4x - 8y$
(1 mark)	$\dfrac{dy}{dx} = \dfrac{-9x^2 + 4x - 8y}{8x}$
(1 mark)	$\left(\dfrac{dy}{dx}\right)_{at\ (1,2)} = \dfrac{-9x^2 + 4x - 8y}{8x}$
(1 mark)	$\left(\dfrac{dy}{dx}\right)_{at\ (1,2)} = \dfrac{-21}{8}$
	$\therefore$ The equation of the tangent line to the graph of function at $(1, 2)$ is
(1 mark)	$y = \left(\dfrac{dy}{dx}\right)_{at\ (1,2)} \times (x - 1) + f(1)$
(1 mark)	$y = \dfrac{-21x + 37}{8}$

Answer Key

19. For $f(x) = x^3 - 27x + 10$, find the intervals where the function is increasing/decreasing, identify the local extreme values, concave up/ concave down, and the inflection point.

Solution:

	Given function is $f(x) = x^3 - 27x + 10 - (1)$
	Differentiating (1) with respect to x on both sides
(0.5 marks)	$f'(x) = 3x^2 - 27$
	$f'(x) = 0 \implies 3x^2 - 27 = 0$ then we get
(1 mark)	$x = \pm 3$ are the critical points
(0.5 marks)	$f(x)$ is increasing in the intervals $(-\infty, -3)$ and $(3, \infty)$
(0.5 marks)	$f(x)$ is decreasing in the interval $(-3, 3)$
(0.5 marks)	at $x = -3$, local maximum occurs for the given function
(0.5 marks)	at $x = 3$, local minimum occurs for the given function
(0.5 marks)	$\because f'(x) = 3x^2 - 27$ then $f''(x) = 6x = 0 \implies x = 0$
(0.5 marks)	$f(x)$ is concave up in the interval $(0, \infty)$
(0.5 marks)	$f(x)$ is concave down in the interval $(-\infty, 0)$
(1 mark)	The inflection of point is $(0, 10)$

Answer Key